Aldrin Kabya Biswas

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Research Interests

Parameter-efficient fine-tuning of language models, activation steering and social bias in quantized LLMs, and vision-language alignment for multimodal learning.

Education

Independent University, Bangladesh (IUB) Dhaka, Bangladesh

B.Sc. in Computer Science & Engineering (CSE) *2020 – 2025*

- **CGPA: 3.88/4.00** (top 1% in CSE; Magna Cum Laude)
- Thesis: *Training-Free Layer Selection for Parameter-Efficient Fine-Tuning of Language Models*
- Supervisors: Amin Ahsan Ali, AKM Mahbubur Rahman
- Relevant coursework: Machine Learning, Artificial Intelligence, Linear Algebra, Numerical Methods, Ordinary Differential Equations

Publications

Journal and Conference Papers

- **Aldrin Kabya Biswas**, Md Fahim, Md Tahmid Hasan Fuad, Akm Moshiur Rahman Mazumder, Amin Ahsan Ali, and AKM Mahbubur Rahman. "Training-Free Layer Selection for Partial Fine-Tuning of Language Models." *Information Sciences* (Q1), 2026. doi:10.1016/j.ins.2026.123204
- **Aldrin Kabya Biswas**, Md Fahim, Md. Ashraful Amin, Amin Ahsan Ali, and AKM Mahbubur Rahman. "PRiSM: Partial Ranking via Inter-layer Semantic Measurement for Efficient Fine-tuning of Language Models." In *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026).* doi:10.63317/3eyz8rr5qun6

Under Review

- **Aldrin Kabya Biswas**, Md. Zakir Hossan, and AKM Mahbubur Rahman. "Disentangling Debiasing from Abstention in Bias-Steering Transfer to Quantized Language Models." Under review at *ACIIDS 2027*.
- Armun Alam, Farhan Mahmud, **Aldrin Kabya Biswas**, and AKM Mahbubur Rahman. "MLLM-Guided Vision-Language Alignment for Land Use Land Cover Segmentation in Remote Sensing." Under review.

Research Experience

Center for Computational & Data Sciences (CCDS), IUB Dhaka, Bangladesh

Research Assistant *September 2025 – Present*

- Generalized a training-free layer selection method for partial fine-tuning to generation tasks; ranking layers by representation similarity in one forward pass, it outperformed existing selection methods on most benchmarks while updating a quarter of the parameters (*Information Sciences*, 2026).
- Investigated whether debiasing steering vectors from full-precision LLMs transfer to quantized models, showing that much of the bias reduction stems from increased abstention rather than less stereotyped answers.
- Built a full-stack WebGIS platform for nationwide land-use and land-cover maps of Bangladesh, with pixel-level change detection between 2019 and 2023 and area statistics for user-defined regions.

Undergraduate Researcher *January 2024 – August 2025*

- Developed and evaluated a layer selection approach that scores layers before training by the cosine similarity of their [CLS] representations, approaching full fine-tuning accuracy on BERT, RoBERTa and DeBERTa (LREC 2026).

Honors & Awards

100% University Merit Scholarship, for outstanding result on the admission test, IUB 2020

Vice Chancellor's Honors List, IUB: Autumn 2021, Spring 2022, Summer 2023, Autumn 2023 2021 – 2023

Dean's Honors List, IUB: Summer 2021, Spring 2023 2021 – 2023

Dean's List, IUB: Summer 2020, Spring 2021, Autumn 2022 2020 – 2022

Presentations

Poster presentation of *PRiSM*, LREC 2026 (virtual) 2026

Three-Minute Thesis (3MT) Competition, IUB 2024

Summer Schools & Workshops

1st CCDS Summer School on Random Matrix Theory 2024

Workshop on Hands-on Deep Learning Coding and Code Management, CCDS 2024

Technical Skills

Languages: Python, C++, JavaScript
Machine Learning: PyTorch, Hugging Face Transformers, bitsandbytes, GPTQModel, HQQ, scikit-learn, SciPy
Web & Geospatial: FastAPI, Next.js, React, GDAL, Rasterio, TiTiler, MapLibre GL JS
Tools: Linux, Git, LaTeX